Nehil Chandrakar

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Technical Skills & Tool Familiarity

Languages: Python, Go, TypeScript, Java
Backend & Systems: Go (net/http, sync.Pool), FastAPI, Node.js, PostgreSQL, MongoDB, Redis, REST APIs
AI Systems: LangGraph, RAG Architectures, Vector Search (ChromaDB, RVSS), ONNX Runtime, Cross-Encoders
Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Docker Compose, GitHub Actions (CI/CD), Git, Linux
CS Fundamentals: Data Structures & Algorithms, System Design, Object-Oriented Programming (OOP), DBMS, Computer Networks

Experience

Alcovia | *Fullstack Engineering Intern* alcovia.life | Jan 2026 – Jun 2026

- **Redesigned and launched** the flagship web platform (alcovia.life) end to end with a custom design system and responsive UI, scaling daily traffic from **4–5 to 1,500–2,000+ visitors** with **sub-second page loads**.
- **Spearheaded** the engineering of Alcovia's core product ecosystem from ground zero, shipping **3 React Native apps**, **2 Next.js web applications**, and a **WebSocket service** backed by a **120+ model PostgreSQL schema** and **400+ API endpoints**.
- **Architected a transactional outbox pipeline** using **PostgreSQL advisory locking**, 5-stage exponential backoff, and a 6-hour reconciler, eliminating race conditions and ensuring **zero data loss** across **23 synchronized ClickUp entity types**.
- **Built an asynchronous Gemini AI extraction pipeline** for session notes and study plans, decoupling LLM workflows from HTTP request paths to eliminate **100% of user-facing timeouts** under production traffic.

Projects

ZenFit: Distributed MultiModal AI Coach | *Go, Python, Redis VSS, RabbitMQ, Docker* | | Docs

- **High-Throughput Go Gateway & Audio Engine:** Engineered a Go API Gateway handling **3,819 req/s** (P50: 7.54ms) at **0.95MB RAM**; built a Go speech microservice streaming raw binary audio at **2,918 req/s** (P50: 28.4ms) with `sync.Pool` zero-alloc buffer reuse, cutting container footprint by **96.9% (520MB → 16.2MB)** and eliminating 33% Base64 overhead.
- **Semantic VSS Caching & Hybrid RRF:** Built an Agentic Self-RAG engine delivering **3.16ms P50 TTFT** (234.5 req/s) via Redis VSS HNSW semantic caching ($D \leq 0.08$), saving 150–250ms of vector compute; fused Dense + BM25 search via Reciprocal Rank Fusion ($k = 60$) with deduplicated parent-child chunk hydration.
- **In-Process ONNX Multilingual Reranking:** Replaced remote API calls with a CPU-native FlashRank ONNX cross-encoder (`ms-marco-MultiBERT-L-12`, <1ms latency) with calibrated confidence gating ($\tau \geq 0.65$), preventing 2.5GB+ Docker image bloat within an 80MB memory footprint.
- **Adaptive Re-Retrieval Ladder:** Deployed a bounded 3-round ladder with MeSH clinical taxonomy shifts, automated plateau circuit breakers ($\Delta < 0.05$) triggering PubMed/DuckDuckGo web fallback, and zero-leak dynamic context isolation ensuring strict biometrics privacy.
- **Sentence-Boundary Audio Streaming:** Pipelined dual-stream HTTP/2 SSE outputs with agentic tool calling for non-blocking UI shimmer cards and a delimiter-aware sentence buffer streaming clauses to a Go Speech service for sub-500ms TTFA.
- **Distributed Async Workers & DLQ Resilience:** Scaled asynchronous plan synthesis via RabbitMQ at **318.4 req/s** (HTTP 202 in <65ms) with distributed Redis idempotency locks, strict post-commit ACKs, poison-pill DLQs, and non-blocking `SCAN + UNLINK` cache invalidation.
- **Zero-Trust Perimeter & Asymmetric RS256 JWKS:** Implemented asymmetric RS256 JWT auth with **RFC 7517 dynamic JWKS discovery**, 15-minute access tokens with rotating cookies, instant Redis denylist session revocation, and zero-host port exposure across 10 Docker containers.

Achievements & Certifications

Mercuria India Hackathon 2024 – 2nd Prize among 450+ teams Nationwide Nov 2024

Emerging Tech Hackathon 2024 – 3rd Place among 200+ professionals & students Dec 2024

Education

Siddaganga Institute of Technology Tumkur, India
B.E. in Computer Science and Engineering (AI/ML) | CGPA: 7.72 *Jun 2026*